

Renesas LPWA Power Estimator

User's Manual

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(Rev.5.0-1 October 2020)

Corporate Headquarters

TOYOSU FORESIA, 3-2-24 Toyosu,
Koto-ku, Tokyo 135-0061, Japan

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General Precautions in the Handling of Microprocessing Unit and Microcontroller Unit Products

The following usage notes are applicable to all Microprocessing unit and Microcontroller unit products from Renesas. For detailed usage notes on the products covered by this document, refer to the relevant sections of the document as well as any technical updates that have been issued for the products.

1. Precaution against Electrostatic Discharge (ESD)

A strong electrical field, when exposed to a CMOS device, can cause destruction of the gate oxide and ultimately degrade the device operation. Steps must be taken to stop the generation of static electricity as much as possible, and quickly dissipate it when it occurs. Environmental control must be adequate. When it is dry, a humidifier should be used. This is recommended to avoid using insulators that can easily build up static electricity. Semiconductor devices must be stored and transported in an anti-static container, static shielding bag or conductive material. All test and measurement tools including work benches and floors must be grounded. The operator must also be grounded using a wrist strap. Semiconductor devices must not be touched with bare hands. Similar precautions must be taken for printed circuit boards with mounted semiconductor devices.

2. Processing at power-on

The state of the product is undefined at the time when power is supplied. The states of internal circuits in the LSI are indeterminate and the states of register settings and pins are undefined at the time when power is supplied. In a finished product where the reset signal is applied to the external reset pin, the states of pins are not guaranteed from the time when power is supplied until the reset process is completed. In a similar way, the states of pins in a product that is reset by an on-chip power-on reset function are not guaranteed from the time when power is supplied until the power reaches the level at which resetting is specified.

3. Input of signal during power-off state

Do not input signals or an I/O pull-up power supply while the device is powered off. The current injection that results from input of such a signal or I/O pull-up power supply may cause malfunction and the abnormal current that passes in the device at this time may cause degradation of internal elements. Follow the guideline for input signal during power-off state as described in your product documentation.

4. Handling of unused pins

Handle unused pins in accordance with the directions given under handling of unused pins in the manual. The input pins of CMOS products are generally in the high-impedance state. In operation with an unused pin in the open-circuit state, extra electromagnetic noise is induced in the vicinity of the LSI, an associated shoot-through current flows internally, and malfunctions occur due to the false recognition of the pin state as an input signal become possible.

5. Clock signals

After applying a reset, only release the reset line after the operating clock signal becomes stable. When switching the clock signal during program execution, wait until the target clock signal is stabilized. When the clock signal is generated with an external resonator or from an external oscillator during a reset, ensure that the reset line is only released after full stabilization of the clock signal. Additionally, when switching to a clock signal produced with an external resonator or by an external oscillator while program execution is in progress, wait until the target clock signal is stable.

6. Voltage application waveform at input pin

Waveform distortion due to input noise or a reflected wave may cause malfunction. If the input of the CMOS device stays in the area between V_{IL} (Max.) and V_{IH} (Min.) due to noise, for example, the device may malfunction. Take care to prevent chattering noise from entering the device when the input level is fixed, and also in the transition period when the input level passes through the area between V_{IL} (Max.) and V_{IH} (Min.).

7. Prohibition of access to reserved addresses

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1. Getting Started

- Set up the operating conditions:
 - Select the device type of the MCU.
 - Select an operating frequency for the MCU.
 - Enter the battery capacity [mAh].
 - Enter the battery self-discharge rate [% / year].
 - Enter the transaction pattern period [ms].
- Create the transaction patterns:
 - Select the operating state of the RF and MCU.
 - Enter the duration [ms] for the above states.
 - Repeat step 2 until end of the transactions.

Finally, you can see the battery life and average current in the **Battery Life** field.

Note: Do not change any cell other than the orange ones.

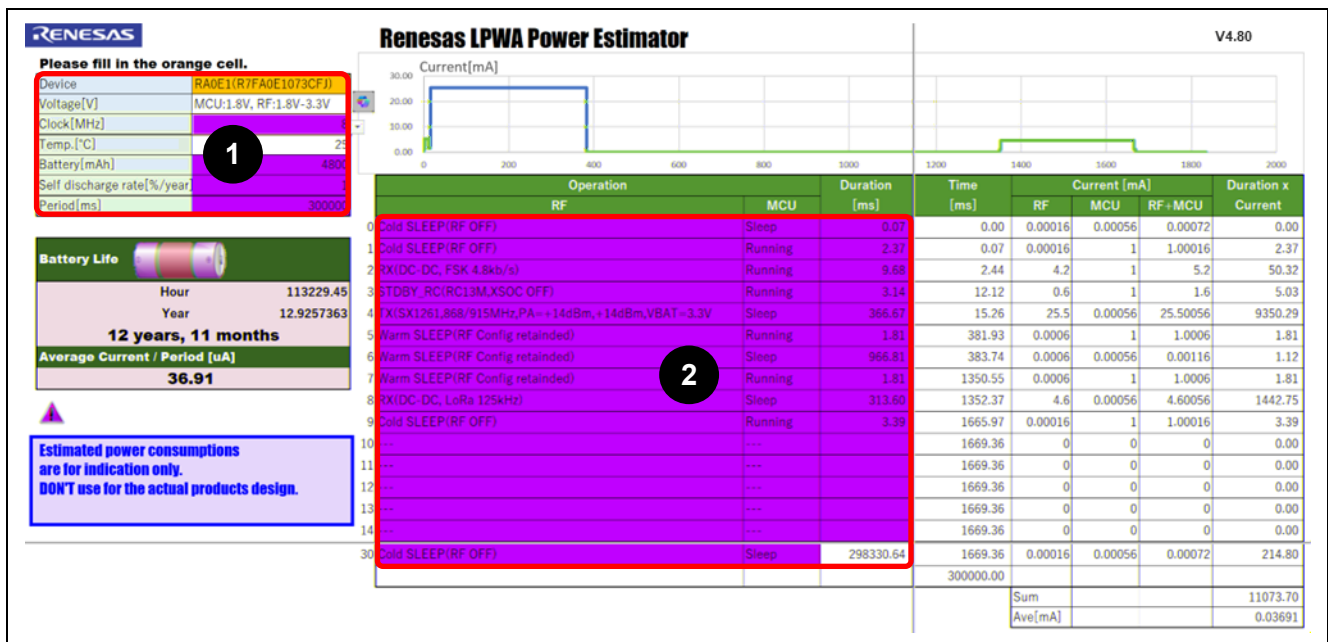


Figure 1. Renesas LPWA Power Estimator GUI

2. Limitations

1. Operating temperature is currently supported only at 25 °C.
2. It is conceivable that the modulation method, RF parameters (SF, CR and so forth) are influenced, but this is not taken into consideration at present.
3. In actual devices, the current during transmission increases with the decrease of the power supply voltage, but this is not considered at present.

Revision History

Rev.	Date	Description	
		Page	Summary
0.15	Mar.23.18	--	1st edition
0.16	July.16.18	3	Change Device name to RL78/G14. Change Self-discharge current to Self-discharge rate. Change Battery from 1500 [mAh] to 4800 [mAh]. Change Period from 3600000 [ms] to 300000 [ms].
1.00	Oct.02.18	--	Change document revision.
1.10	May.23.19	3	Change target device from R5F104JJ to R5F104ML/R5F104GL.
1.11	Sep.26.19	All	Change program name.
3.00	Mar.02.21	All	Change program name.
		All	Supports RL78/G23(R7F100GLG, 64 pin/128 KB).
3.10	Sep.20.21	All	Supports RL78/G23(R7F100GSN, 128 pin/768 KB).
4.00	Aug.29.22	All	Supports RA2E1(R7FA2E1A9xxFM, 64 pin/128 KB).
4.20	Mar.31.23	All	Supports RA2L1(R7FA2L1ABxxFP, 100 pin/256 KB)
4.30	Apr.21.23	All	Supports RL78/G22(R7F102GGE, 48 pin/64 KB)
4.50	May.24.24	All	Supports RA0E1(R7FA0E1073CFJ, 32 pin/64 KB)
4.70	Apr.18.25	All	Supports RA0E2(R7FA0E2094CFM, 64pin/128KB)
4.80	Aug.21.25	All	Supports RL78/L23(R7F100LPL3CFB, 100pin, 512-KB ROM (256 KB x 2 banks), 32-KB RAM)

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Publication Date: Aug.21.25

Published by: Renesas Electronics Corporation

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